PART IIA, PAPER 3, SUPERVISION 2
OLEG I. KITOV, UNIVERSITY OF CAMBRIDGE

SOLUTIONS TO PROBLEMS

S2.1 Concave millennials return to education.

In order to understand the determinants of years of schooling among women in the UK, we collected a random sample of $n = 857$ individual females aged 28-38 from across the UK and gathered information on a number of background characteristics including their score on an IQ test. The summary statistics are as follows.

Table 1: Summary statistics for explanatory variables

Variable	Mean	Std. Dev.	Min	Max
`educ`	13.58	2.20	9	18
`age`	32.98	3.09	28	38
`black`	0.12	0.32	0	1
`meduc`	10.68	2.85	0	18
`IQ`	101.80	15.01	50	145

The variable `educ` represents years of schooling, `age` is recorded in years and `agesq` is the square of `age`, `meduc` is the number of years of schooling for the individual's mother, `IQ` is the IQ test score and `black` is a dummy variable for the African origins. Define the log of years of education as $\texttt{leduc} = \ln \texttt{educ}$. A linear regression of `leduc` on the explanatory variables from Table 1 is estimated using OLS. The corresponding coefficient sample estimates of $\hat{\beta}_j$ are reported in Table 2 below, together with their standard errors $s_{\hat{\beta}_j}$, and the corresponding 2-sided p-values for the hypothesis test of $H_0\colon \beta_j = 0$ against $H_1\colon \beta_j \neq 0$. For this regression, we have $R^2 = 0.33$.

Table 2: OLS regression for `leduc` as the dependent variable

Regressor	Coeff. $\hat{\beta}$	Std. error $s_{\hat{\beta}}$	p-value
`IQ`	0.0050	0.0003	0.000
`age`	0.1202	0.0361	0.001
`agesq`	−0.0018	0.0005	0.001
`black`	−0.0314	0.0152	0.039
`meduc`	0.0126	0.0017	0.000
`constant`	−0.0569	0.5981	0.924

(a) Interpret the coefficient 0.005 on `IQ`.

Solution: First, write down model specification:

$$\ln \texttt{educ}_i = \beta_0 + \beta_1 \texttt{IQ}_i + \beta_2 \texttt{age}_i + \beta_3 \texttt{age}_i^2 + \beta_4 \texttt{black}_i + \beta_5 \texttt{meduc}_i + u_i.$$

The OLS estimator of β_1 is the partial effect of IQ on log education, so:

$$\left.\frac{\partial \ln \texttt{educ}_i}{\partial \texttt{IQ}_i}\right|_{\texttt{age,black,meduc}} = \left(\frac{\partial \texttt{educ}_i}{\texttt{educ}_i}\right)\left(\frac{1}{\partial \texttt{IQ}_i}\right) = \beta_1$$

A one point increase in IQ raises education by 0.5%, holding age, race and mother's education constant.

(b) Compute the partial effect of age on `leduc`. Sketch and interpret the relationship between `leduc` and age. What does the partial effect represent in your plot?

Solution: Differentiate $\ln \texttt{educ}_i$ with respect to$\texttt{age}_i$:

$$\frac{\partial \ln \texttt{educ}_i}{\partial \texttt{age}_i} = \beta_2 + 2\beta_3 \texttt{age}_i.$$

Hence, partial effect of age depends on age. Since $\hat{\beta}_2 = 0.1202 > 0$ and $\hat{\beta}_3 = -0.0018 < 0$, the relationship between log education and age is concave. Note that the age for which the partial effect of age on log education becomes negative is 33.38.

(c) How would you test whether educational attainment is affected by age?

Solution: Since we have two explanatory variables that depend on age, we want to test joint hypothesis for $q = 2$ restrictions, $H_0\colon\ (\beta_2 = 0) \cap (\beta_3 = 0)$ against $H_1\colon\ (\beta_2 \neq 0) \cup (\beta_3 \neq 0)$. To conduct the test, estimate the restricted model:

$$\ln \texttt{educ}_i = \pi_0 + \pi_1 \texttt{IQ}_i + \pi_4 \texttt{black}_i + \pi_5 \texttt{meduc}_i + v_i$$

and use the F-test statistic that follows a $F_{2,n-6}$ distribution:

$$F = \frac{(SSR_r - SSR_u)/2}{SSR_u/(n-6)} = \frac{\left(R_u^2 - R_r^2\right)/2}{\left(1 - R_u^2\right)/(n-6)} \sim F_{q,n-k-1}$$

where SSR_r $\left(R_r^2\right)$ and SSR_u $\left(R_u^2\right)$ are the sums of squared residuals (coefficient of determination) from the restricted the the unrestricted model, respectively.

(d) How would the coefficients and the intercept change if age was instead recorded in months? What would happen to the corresponding t-statistics? Explain.

Solution: Suppose that the new variable $\texttt{age}_i^* = 12\texttt{age}_i$ is now used in the original regression instead of $\texttt{age}_i$. Consequently, the variable $\texttt{age}_i^2$ is now replaced with $\texttt{age}_i^* = 144\texttt{age}_i$. The new regression is

$$\begin{aligned}
\ln \texttt{educ}_i &= \pi_0 + \pi_1 \texttt{IQ}_i + \pi_2 \texttt{age}_i^* + \pi_3 \left(\texttt{age}_i^*\right)^2 + \pi_4 \texttt{black}_i + \pi_5 \texttt{meduc}_i + u_i \\
&= \pi_0 + \pi_1 \texttt{IQ}_i + \pi_2 \left(12\texttt{age}_i\right) + \pi_3 \left(144\texttt{age}_i^2\right) + \pi_4 \texttt{black}_i + \pi_5 \texttt{meduc}_i + u_i \\
&= \pi_0 + \pi_1 \texttt{IQ}_i + 12\pi_2 \texttt{age}_i + 144\pi_3 \texttt{age}_i^2 + \pi_4 \texttt{black}_i + \pi_5 \texttt{meduc}_i + u_i \\
&= \beta_0 + \beta_1 \texttt{IQ}_i + \beta_2 \texttt{age}_i + \beta_3 \texttt{age}_i^2 + \beta_4 \texttt{black}_i + \beta_5 \texttt{meduc}_i + u_i
\end{aligned}$$

The intercept does not change. However, since the transformations of the variable do not affect the partial effects of age on education, the corresponding coefficients would change proportionally, so that $\pi_2 = \beta_2/12$ and $\pi_3 = \beta_3/144$. The standard errors

$s_{\hat{\pi}_2}$ and $s_{\hat{\pi}_3}$ would increase proportionally by a factor of 12 and 144, respectively, since

$$\begin{aligned} s^2_{\hat{\pi}_2} = \widehat{\mathrm{Var}}\left(\hat{\pi}_2\right) &= \frac{\hat{\sigma}^2}{\sum_{i=1}^n \left(\mathtt{age}_i^* - \overline{\mathtt{age}^*}\right)^2 \left(1 - R^2_{\mathtt{age}^*}\right)} \\ &= \frac{\hat{\sigma}^2}{144 \sum_{i=1}^n \left(\mathtt{age}_i - \overline{\mathtt{age}}\right)^2 \left(1 - R^2_{\mathtt{age}}\right)} \\ &= \left(\frac{1}{12}\right)^2 \widehat{\mathrm{Var}}\left(\hat{\beta}_2\right). \end{aligned}$$

where $\hat{\sigma}^2$ is the estimator of error variance, and $R^2_{\mathtt{age}^*}$ is the coefficient of determination from the regression of $\mathtt{age}_i^*$ on all the other explanatory variables. Clearly, $R^2_{\mathtt{age}^*} = R^2_{\mathtt{age}}$. Consequently, the t-statistics for $\hat{\beta}_2$ would not change

$$t_{\hat{\pi}_2} = \frac{\hat{\pi}_2}{s_{\hat{\pi}_2}/\sqrt{n}} = \frac{\hat{\beta}_2/12}{\frac{1}{12} s_{\hat{\beta}_2}/\sqrt{n}} = t_{\hat{\beta}_2}.$$

This makes sense, since statistical significance of the explanatory variable should not be affect by measurement units. The same argument applies to $\hat{\pi}_3$.

(e) How would you test that the true coefficient on mothers education is 1/3 the negative of the true coefficient on black?

Solution: We would like to test $H_0\colon \beta_5 = -1/3\beta_4$ or alternatively that $3\beta_5 + \beta_4 = 0$. Reparametrize the model by adding and subtracting $3\beta_5\mathtt{black}_i$ in the original regression:

$$\begin{aligned} \ln \mathtt{educ}_i &= ... + \beta_4\mathtt{black}_i + 3\beta_5\mathtt{black}_i + \beta_5\mathtt{meduc}_i - 3\beta_5\mathtt{black}_i + u_i \\ &= ... + \left(\beta_4 + 3\beta_5\right)\mathtt{black}_i + \beta_5\left(\mathtt{meduc}_i - 3\mathtt{black}_i\right) + u_i \end{aligned}$$

Define a new variable $z_i = \mathtt{meduc}_i - 3\mathtt{black}_i$ and regress:

$$\ln \mathtt{educ}_i = \theta_0 + \theta_1\mathtt{IQ}_i + \theta_2\mathtt{age}_i + \theta_3\mathtt{age}_i^2 + \theta_4\mathtt{black}_i + \theta_5 z_i + u_i$$

In this new specification, testing $H_0\colon \theta_4 = 0$ would be equivalent to testing $H_0\colon \beta_5 = -1/3\beta_4$ in the original regression. In `STATA` could simply do

```
>> reg lwage educ age agesq black meduc
>> test (3*meduc = -black)
```

S2.2 Best in Show (of the linear unbiased estimators breed).

This question is about applying the law of iterated expectations (LIE) [see Supervision 1 for details on the LIE] to compute the expectation and the variance of coefficient estimators in a linear regression when the regressors are not fixed numbers, but are instead random variables. Consider the following population regression

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \ldots, n. \tag{1}$$

Suppose the error terms u_i are iid for $i = 1, \ldots, n$, such that $\mathbb{E}\left[u_i|\mathbf{X}\right] = 0$ and $\mathrm{Var}\left(u_i|\mathbf{X}\right) = \sigma^2$. Denote the vector of observations on the explanatory variable by

$\mathbf{X} = (X_1, \ldots, X_n)^\intercal$. Here, we introduce arguably the most important assumption of the year: explanatory variables X_i are **NOT fixed numbers**, but rather they are **random variables**. This is a relaxation of the overly-simplifying assumption (A2) in the Gauss-Markov theorem, which we've gotten so used to. Now, it's time to drop this in favour of a more realistic setup. It is integral to stress one more time that X_i are no longer fixed (non-random) numbers. Instead, X_i are assumed to be random and are drawn from some distribution. Hence, X_i have an expectation and a variance. Most importantly, since X_i is now a random variable and not a fixed number, $\mathbb{E}[X_i] \neq X_i$. However, it is true that $\mathbb{E}[X_i|X_i = x] = x$, or alternatively $\mathbb{E}[X_i|X_i] = X_i$. This somewhat tautological statement says that conditional on the knowledge that X_i takes a realization x, the expectation of X_i is in fact x. Now, let's get back to regression (1) and let $\hat{\beta}_1$ denote the OLS estimator of β_1. Consider another linear estimator $\tilde{\beta}_1$ of β_1, constructed in the following way, assuming $X_3 + X_1 - 2X_2 \neq 0$:

$$\tilde{\beta}_1 = \frac{Y_3 + Y_1 - 2Y_2}{X_3 + X_1 - 2X_2}. \tag{2}$$

(a) Express $\tilde{\beta}_1$ in terms of β_1, X_1, X_2, X_3 and u_1, u_2, u_3.
Solution: Substitute for Y_1, Y_2, Y_3 from (1) into (2)

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\beta_0 + \beta_1 X_3 + u_3 + \beta_0 + \beta_1 X_1 + u_1 - 2\beta_0 - 2\beta_1 X_2 - 2u_2}{X_3 + X_1 - 2X_2} \\
&= \frac{\beta_1 (X_3 + X_1 - 2X_2) + (u_3 + u_1 - 2u_2)}{X_3 + X_1 - 2X_2} \\
&= \beta_1 + \frac{u_3 + u_1 - 2u_2}{X_3 + X_1 - 2X_2}.
\end{aligned}$$

(b) Use your expression in (a) to find $\mathbb{E}\left[\tilde{\beta}_1\right]$. Is $\tilde{\beta}_1$ unbiased for β_1? [Hint: recall that $\mathbb{E}[X_i] \neq X_i$, but $\mathbb{E}[X_i|X_i] = X_i$, and realize that you can find $\mathbb{E}\left[\tilde{\beta}_1\right]$ only by using the law of iterated expectations. First you need to find $\mathbb{E}\left[\tilde{\beta}_1|X_1, X_2, X_3\right]$ and then apply the law of iterated expectations.]
Solution: Since X_1, X_2, X_3 are now random variables, not fixed numbers, we cannot simply take the unconditional expectation of the expression for $\tilde{\beta}_1$ in (a). Instead, we must first compute the conditional expectation $\mathbb{E}\left[\tilde{\beta}_1|X_1, X_2, X_3\right]$ and then invoke the law of iterated expectations. We know that $\mathbb{E}[X|X] = X$ and that $\mathbb{E}[u_i|X_1, X_2, X_3] = 0$ for $i = 1, 2, 3$. Moreover, we can show using the law of iterated expectations that the error terms are also unconditionally mean zero:

$$\mathbb{E}[u_i] = \mathbb{E}[\mathbb{E}[u_i|X_1, X_2, X_3]] = \mathbb{E}[0] = 0, \quad i = 1, 2, 3 \tag{3}$$

Now, let's compute the conditional expectation of $\tilde{\beta}_1$:

$$\begin{aligned}
\mathbb{E}\left[\tilde{\beta}_1|X_1, X_2, X_3\right] &= \mathbb{E}\left[\beta_1 + \frac{u_3 + u_1 - 2u_2}{X_3 + X_1 - 2X_2} \,\middle|\, X_1, X_2, X_3\right] \\
&= \beta_1 + \mathbb{E}\left[\frac{u_3 + u_1 - 2u_2}{X_3 + X_1 - 2X_2} \,\middle|\, X_1, X_2, X_3\right] \\
&= \beta_1 + \left(\frac{1}{X_3 + X_1 - 2X_2}\right) \mathbb{E}[u_3 + u_1 - 2u_2 \,|\, X_1, X_2, X_3] \\
&= \beta_1.
\end{aligned}$$

Here we used the fact that

$$\mathbb{E}\left[u_3+u_1-2u_2|X_1,X_2,X_3\right]=\mathbb{E}\left[u_3|X_1,X_2,X_3\right]+\mathbb{E}\left[u_1|X_1,X_2,X_3\right] \\ -2\mathbb{E}\left[u_2|X_1,X_2,X_3\right]=0$$

Finally, invoke the law of iterated expectations to show that

$$\mathbb{E}\left[\tilde{\beta}_1\right]=\mathbb{E}\left[\mathbb{E}\left[\tilde{\beta}_1|X_1,X_2,X_3\right]\right]=\mathbb{E}\left[\beta_1\right]=\beta_1. \tag{4}$$

(c) Use your expression in (a) to show that the unconditional variance of $\tilde{\beta}_1$ is given by

$$\mathrm{Var}\left(\tilde{\beta}_1\right)=6\sigma^2\mathbb{E}\left[\frac{1}{\left(X_3+X_1-2X_2\right)^2}\right]. \tag{5}$$

Do you think that $\tilde{\beta}_1$ can be a better estimator than $\hat{\beta}_1$? [Hint: first, find the conditional variance $\mathrm{Var}\left(\tilde{\beta}_1|X_1,X_2,X_3\right)$ using the first principles and then apply the law of iterated expectations.]

Solution: First note that since β_1 is a constant parameter,

$$\mathrm{Var}\left(\tilde{\beta}_1\right)=\mathrm{Var}\left(\beta_1+\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right)=\mathrm{Var}\left(\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right).$$

Now, we can use the formula for the variance of a random variable Y, $\mathrm{Var}\left(Y\right)=\mathbb{E}\left[Y^2\right]-\left(\mathbb{E}\left[Y\right]\right)^2$, applied to the right-hand side of the above expression to show that

$$\begin{aligned}\mathrm{Var}\left(\tilde{\beta}_1\right)&=\mathrm{Var}\left(\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right)\\&=\mathbb{E}\left[\left(\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right)^2\right]-\left(\mathbb{E}\left[\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right]\right)^2\\&=\mathbb{E}\left[\left(\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right)^2\right].\end{aligned}$$

The second term in the second line is canceled out since $\mathbb{E}\left[u_i\right]=0$, $i=1,2,3$. First, expand the square:

$$\mathbb{E}\left[\left(\frac{u_3+u_1-2u_2}{X_3+X_1-2X_2}\right)^2\right]=\mathbb{E}\left[\frac{u_3^2+u_1^2+4u_2^2+2u_3u_1-4u_3u_2-4u_1u_2}{\left(X_3+X_1-2X_2\right)^2}\right].$$

Then, take conditional expectation of this term

$$\mathbb{E}\left[\frac{u_3^2+u_1^2+4u_2^2+2u_3u_1-4u_3u_2-4u_1u_2}{\left(X_3+X_1-2X_2\right)^2}\Big|X_1,X_2,X_3\right].$$

The denominator can be taken out of the expectation term since

$$\mathbb{E}\left[\frac{1}{\left(X_3+X_1-2X_2\right)^2}\Big|X_1,X_2,X_3\right]=\frac{1}{\left(X_3+X_1-2X_2\right)^2}.$$

The components of the numerator can be considered separately, since the expectation operator is linear. Use the fact that $\mathrm{Var}\left(u_i|\mathbf{X}\right) = \sigma^2$, $i = 1, \ldots, n$, to simplify the square error terms:

$$\begin{aligned}
\mathbb{E}\left[u_3^2 \,|\, X_1, X_2, X_3\right] &= \sigma^2 \\
\mathbb{E}\left[u_1^2 \,|\, X_1, X_2, X_3\right] &= \sigma^2 \\
\mathbb{E}\left[4u_2^2 \,|\, X_1, X_2, X_3\right] &= 4\sigma^2
\end{aligned}$$

Then, use the fact that u_i are independent and so for any $i \neq j$

$$\mathrm{Cov}\left(u_i, u_j\right) = \mathbb{E}\left[u_i u_j\right] - \mathbb{E}\left[u_i\right]\mathbb{E}\left[u_j\right] = \mathbb{E}\left[u_i u_j\right] = 0.$$

Use this to simplify the cross terms in the expectation:

$$\begin{aligned}
\mathbb{E}\left[2u_3u_1 \,|\, X_1, X_2, X_3\right] &= 0 \\
\mathbb{E}\left[-4u_3u_2 \,|\, X_1, X_2, X_3\right] &= 0 \\
\mathbb{E}\left[-4u_1u_2 \,|\, X_1, X_2, X_3\right] &= 0
\end{aligned}$$

Combine all of these results to conclude that the conditional variance of $\tilde{\beta}_1$ is

$$\mathrm{Var}\left(\tilde{\beta}_1|X_1, X_2, X_3\right) = \mathbb{E}\left[\left(\frac{u_3 + u_1 - 2u_2}{X_3 + X_1 - 2X_2}\right)^2 \Big| X_1, X_2, X_3\right] = \frac{6\sigma^2}{\left(X_3 + X_1 - 2X_2\right)^2}.$$

Finally, invoke the law of iterated expectations to find the unconditional variance:

$$\begin{aligned}
\mathrm{Var}\left(\tilde{\beta}_1\right) &= \mathbb{E}\left[\mathbb{E}\left[\left(\frac{u_3 + u_1 - 2u_2}{X_3 + X_1 - 2X_2}\right)^2 \Big| X_1, X_2, X_3\right]\right] \\
&= \mathbb{E}\left[\frac{6\sigma^2}{\left(X_3 + X_1 - 2X_2\right)^2}\right] \\
&= 6\sigma^2\mathbb{E}\left[\frac{1}{\left(X_3 + X_1 - 2X_2\right)^2}\right].
\end{aligned}$$

A crucial thing to note is that this expression cannot be simplified further since

$$\mathbb{E}\left[\frac{1}{\left(X_3 + X_1 - 2X_2\right)^2}\right] \neq \frac{1}{\left(X_3 + X_1 - 2X_2\right)^2}.$$

But we can condition on X_1, X_2, X_3 and get an expression for the conditional variance of the estimator:

$$\mathrm{Var}\left(\tilde{\beta}_1|X_1, X_2, X_3\right) = 6\sigma^2\mathbb{E}\left[\frac{1}{\left(X_3 + X_1 - 2X_2\right)^2}|X_1, X_2, X_3\right] = \frac{6\sigma^2}{\left(X_3 + X_1 - 2X_2\right)^2}$$

(d) Consider a generalized version of the $\tilde{\beta}_1$ estimator, expressed as

$$\beta_1^* = \left(\sum_{i=1}^n \omega_i Y_i\right)\left(\sum_{i=1}^n \omega_i X_i\right)^{-1} \tag{6}$$

where ω_i are weights assigned by the estimator to each observation (Y_i, X_i), for $i = 1, \ldots, n$. Under what condition on ω_i is β_1^* unbiased for β_1? Note that under this condition β_1^* defines a whole class of linear unbiased estimators. Use (5) to deduce an expression for $\mathrm{Var}(\beta_1^*)$. Can β_1^* be a better estimator than $\hat{\beta}_1$?

Solution: From (a) and (b) it is clear that the condition for unbiasedness is that all weights sum up to zero $\sum_{i=1}^n \omega_i = 0$. Based on (5) it is easy to deduce that

$$\mathrm{Var}\left(\beta_1^*\right) = \left(\sigma^2 \sum_{i=1}^n \omega_i^2\right) \mathbb{E}\left[\left(\sum_{i=1}^n \omega_i X_i\right)^2\right]. \tag{7}$$

and consequently the conditional variance is

$$\mathrm{Var}\left(\beta_1^* | \mathbf{X}\right) = \frac{\sigma^2 \sum_{i=1}^n \omega_i^2}{\left(\sum_{i=1}^n \omega_i X_i\right)^2}.$$

β_1^* cannot be a better estimator than $\hat{\beta}_1$, since by Gauss-Markov theorem OLS is best, it has the lowest variance among all unbiased linear estimators, which are indeed represented by β_1^*.

(e) Show that the OLS estimator $\hat{\beta}_1$ is a special case of the general linear unbiased estimator β_1^*. In other words, write down an expression for the weights ω_i in (6) that correspond to the OLS estimator. Finally, explain what makes OLS weights so special and hence OLS the "best" linear unbiased estimator.

Solution: Recall the OLS estimator can be written as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) Y_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right) X_i} = \frac{\sum_{i=1}^n \omega_i^{\mathrm{OLS}} Y_i}{\sum_{i=1}^n \omega_i^{\mathrm{OLS}} X_i}$$

where the weights are given by the deviation of the regressor from its mean

$$\omega_i^{\mathrm{OLS}} = \left(X_i - \bar{X}\right).$$

The OLS weights are very special and are in fact best, in a sense that these are the weights that minimize the variance of the general linear unbiased estimator. In fact, one can think of getting the OLS estimators by setting up the estimation problem as minimization of the variance of the general linear unbiased estimator with respect to the weights of individual observations (note that in the optimization problem below, the only constraint is that the weights sum up to zero, which as we know is the requirement for unbiasedness):

$$\min_{\omega_1, \ldots, \omega_n} \mathrm{Var}\left(\beta_1^* | \mathbf{X}\right) = \min_{\omega_1, \ldots, \omega_n} \left[\frac{\sigma^2 \sum_{i=1}^n \omega_i^2}{\left(\sum_{i=1}^n \omega_i X_i\right)^2}\right]$$
$$\text{s.t.} \quad \sum_{i=1}^n \omega_i = 0.$$

S2.3 Don't let the correlated regressors get away with impartiality.

In this question we would like to show explicitly that in a multivariate linear regression, the coefficients are interpreted as partial effects, not absolute effects. Namely,

in a regression of Y_i on k explanatory variables, $X_{i1}, X_{i2}, \ldots, X_{ik}$:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + u_i, \quad i = 1, \ldots, n, \tag{8}$$

the coefficient β_j, $j = 1, \ldots, k$, is interpreted as the effect of X_{ij} on Y_i that cannot be explain by all other variables. We typically say that β_j is the partial effect of X_{ij} on Y_i, or it is the effect of X_{ij} on Y_i keeping other variables fixed (*ceteris paribus*). Formally, the coefficient β_j is given by

$$\beta_j = \frac{\mathrm{Cov}\,(Y_i, \varepsilon_{ij})}{\mathrm{Var}\,(\varepsilon_{ij})}, \tag{9}$$

where ε_{ij} is the error for observation i from the regression of X_{ij} on all other explanatory variables. For example,

$$\beta_1 = \frac{\mathrm{Cov}\,(Y_i, \varepsilon_{i1})}{\mathrm{Var}\,(\varepsilon_{i1})}, \tag{10}$$

where ε_{i1} is the residual for observation i from the following regression:

$$X_{i1} = \delta_0 + \delta_2 X_{i2} + \delta_3 X_{i3} + \cdots + \delta_k X_{ik} + \varepsilon_{i1}. \tag{11}$$

The error term ε_{i1} is then the part of the original variable X_{i1} that cannot be explained by other regressors. Hence, β_1 is the effect that X_{i1} has on Y_i after we take into consideration (partial out) the effects that all other variables have on Y_i through their influence on X_{i1}. The reason for this interpretation steams from the fact that explanatory variables can in principle be, and in reality tend to be, correlated.

Suppose that in regression (8), X_{i1} depends on / is correlated with X_{i2}. Consider what happens in this setting when X_{i2} changes. This change will produce two effects on Y_i. First, a change in X_{i2} will directly affect Y_i in expression (8). Second, a change in X_{i2} will lead to a change in X_{i1} through regression (11), thereby having an indirect effect on Y_i via (8). If we are not aware that a change in X_{i1} was fully determined by a change in X_{i2}, this indirect effect will be masked as the influence of X_{i1} on Y_i. This influence is of course not genuine, since it was a change in X_{i2} alone that triggered the overall effect on Y_i. Consequently, we would like β_1 to capture exclusively the effect of X_{i1} on Y_i that is not a masked by-product of a change in X_{i2} (or any other variables), but is the genuine effect of a change in X_{i1} only.

Intuitively, in order to do that, we need to extract the component of X_{i1} that cannot be explained by any other variables in regression (8). As we have defined above, this component is precisely the error term ε_{i1} from regression (11) of X_{i1} on all other explanatory variables. Furthermore, let's have another close look at the expression for β_1 in (10). It does look surprisingly familiar. In fact, β_1 is the slope coefficient of ε_{i1} in the bivariate population regression of Y_i on ε_{i1}:

$$Y_i = \alpha_1 + \beta_1 \varepsilon_{i1} + v_{i1}. \tag{12}$$

The slope α_1 and residuals v_{i1} in this regression are of no particular interest to us. It is only the slope β_1 that we care about. We can interpret it in the usual way as the effect that a small change in ε_{i1} has on Y_i, formally $\beta_1 = \partial Y_i / \partial \varepsilon_{i1}$. This interpretation is consistent with the argument above. Namely, β_1 captures the effect of a small change in X_{i1} that cannot be explained by changes in other explanatory

variables, i.e. ε_{i1}, on Y_i. In other words, β_1 is the effect that a small change in X_{i1} has on Y_i, provided that no other variables changed (*ceteris paribus*, all other things being equal). Equivalently, we call this the partial effect of X_{i1} on Y_i.

Now that we know the interpretation, we would like to derive the expression for the partial effect in (9) explicitly. In order to do that, consider a simplified version of (8) with only two explanatory variables, X_{i1} and X_{i2}. The population regression is then given by the following specification:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i, \quad i = 1, \ldots, n. \tag{13}$$

Assume that the conditional zero mean assumption holds, $\mathbb{E}[u_i | X_{i1}, X_{i2}] = 0$ for all i. We will show that the coefficient β_1 in this linear regression is given by

$$\beta_1 = \frac{\text{Cov}(Y_i, \varepsilon_{i1})}{\text{Var}(\varepsilon_{i1})}, \tag{14}$$

where ε_{i1} is the error from the regression of X_{i1} on X_{i2}:

$$X_{i1} = \delta_0 + \delta_1 X_{i2} + \varepsilon_{i1}. \tag{15}$$

In this specification the zero conditional mean assumption holds by construction, $\mathbb{E}[\varepsilon_{i1} | X_{i2}] = 0$. Recall that the residual ε_{i1} is the component of X_{i1} that cannot be explained by X_{i2}.

(a) Suppose that for all observations i, the value of X_{i2} changes by ΔX_{i2}, and its new value is now $X_{i2}^* = X_{i2} + \Delta X_{i2}$. Assume that u_i in regression (13) and ε_{i1} in regression (15) remain unchanged. Derive the effects that this change has on X_{i1} and Y_i and express them in terms of ΔX_{i2}. Interpret your expressions.

Solution: We would like to find the new values of X_{i2} and Y_i. Denote these by X_{i2}^* and Y_i^*, respectively. First, consider regression (15). The component ε_{i1} of X_{i1} unaffected by X_{i2} remains the same for all i. Hence, following a change in X_{i2}, the new value of X_{i2} becomes

$$\begin{aligned} X_{i1}^* &= \delta_0 + \delta_1 X_{i2}^* + \varepsilon_{i1} \\ &= \delta_0 + \delta_1 (X_{i2} + \Delta X_{i2}) + \varepsilon_{i1} \\ &= \delta_0 + \delta_1 X_{i2} + \delta_1 \Delta X_{i2} + \varepsilon_{i1} \\ &= X_{1i} + \delta_1 \Delta X_{i2}. \end{aligned}$$

The value of X_{i1} changes by the amount $\delta_1 \Delta X_{i2}$. Now, let's find the effect on Y_i by considering what happens to its value in regression (13). First, observe that u_i is the component of Y_i unaffected by X_{i1} and X_{i2} so it remains unchanged. Then, substitute the values for X_{i1}^* and X_{i2}^* into (13)

$$\begin{aligned} Y_i^* &= \beta_0 + \beta_1 X_{i1}^* + \beta_2 X_{i2}^* + u_i \\ &= \beta_0 + \beta_1 (X_{i1} + \delta_1 \Delta X_{i2}) + \beta_2 (X_{i2} + \Delta X_{i2}) + u_i \\ &= \beta_0 + \beta_1 X_{i1} + \beta_1 \delta_1 \Delta X_{i2} + \beta_2 X_{i2} + \beta_2 \Delta X_{i2} + u_i \\ &= (\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i) + (\beta_1 \delta_1 + \beta_2) \Delta X_{i2} \\ &= Y_i + (\beta_2 + \beta_1 \delta_1) \Delta X_{i2}. \end{aligned}$$

The value of Y_i changes by the amount $(\beta_1\delta_1 + \beta_2)\,\Delta X_{i2}$, which is driven purely by ΔX_{i2}. This little change in X_{i2} propagates through the system of interrelationships and affect all variables: first X_{i1} via (15), and then Y_i via a change in its own value and via the consequential effect it had on X_{i1}. The overall influence of ΔX_{i2} on Y_i can then be decomposed into two components. The first, β_2, is the direct effect of X_{i2} on Y_i. This is our partial effect in the original regression (13), the one we are really interested in when talking about the influence of X_{i2} on Y_i. The second, $\beta_1\delta_1$, is the indirect effect, which is determined jointly by δ_1 - the effect of X_{i2} on X_{i1}, and β_1 - the effect of X_{i1} on Y_i. Since we usually only observe the values of X^*_{i1} and X^*_{i2}, rather than ΔX_{i2}, this indirect effect will be masked by $\beta_1 X^*_{i1}$. It will appear in the estimated regression (13) as the effect of X_{i1} on Y_i, which is not what we are after when interpreting the partial effect of X_{i2} on Y_i. Finally, note that since ε_{i1} remained fixed, the change in X_{i2} must have been driven purely by ε_{i2}, the component of X_{i2} unaffected by X_{i1}.

(b) Show that ε_{i1} is uncorrelated with X_{i2} and u_i.

Solution: The zero conditional mean assumption $\mathbb{E}\left[\varepsilon_{i1}|X_{i2}\right] = 0$ implies that ε_{i1} is uncorrelated with X_{i2}, that is $\mathrm{Cov}\left(\varepsilon_{i1}, X_{i2}\right) = 0$. To show that ε_{i1} is uncorrelated with u_i, first notice that ε_{i1} is a linear combination of the original regressors X_{i1} and X_{i2}, namely

$$\varepsilon_{i1} = X_{i1} - \delta_0 - \delta_1 X_{i2}. \tag{16}$$

Then, using the fact that u_i is uncorrelated with X_{i1} and X_{i2} by the zero conditional mean assumption $\mathrm{E}\left[u_i|X_{i1}, X_{i2}\right] = 0$ from the original regression (13), we get that

$$\mathrm{Cov}\left(\varepsilon_{i1}, u_i\right) = \mathrm{Cov}\left(X_{i1} - \delta_0 - \delta_1 X_{i2}, u_i\right) = \mathrm{Cov}\left(X_{i1}, u_i\right) - \delta_1 \mathrm{Cov}\left(X_{i2}, u_i\right) = 0.$$

(c) Show that $\mathrm{Cov}\left(X_{i1}, \varepsilon_{i1}\right) = \mathrm{Var}\left(\varepsilon_{i1}\right)$.

Solution: Substitute for X_{i1} from (15) into the expression for covariance:

$$\begin{aligned}
\mathrm{Cov}\left(X_{i1}, \varepsilon_{i1}\right) &= \mathrm{Cov}\left(\delta_0 + \delta_1 X_{i2} + \varepsilon_{i1}, \varepsilon_{i1}\right) \\
&= \mathrm{Cov}\left(\delta_0, \varepsilon_{i1}\right) + \mathrm{Cov}\left(\delta_1 X_{i2}, \varepsilon_{i1}\right) + \mathrm{Cov}\left(\varepsilon_{i1}, \varepsilon_{i1}\right) \\
&= \delta_1 \mathrm{Cov}\left(X_{i2}, \varepsilon_{i1}\right) + \mathrm{Var}\left(\varepsilon_{i1}\right) \\
&= \mathrm{Var}\left(\varepsilon_{i1}\right).
\end{aligned}$$

Here we used the fact that $\mathrm{Cov}\left(\varepsilon_{i1}, \varepsilon_{i1}\right) = \mathrm{Var}\left(\varepsilon_{i1}\right)$ and that $\mathrm{Cov}\left(X_{i2}, \varepsilon_{i1}\right) = 0$ by zero conditional mean assumption from (15), that is the residual ε_{i1} is uncorrelated with the regressor X_{i2}.

(d) Show that $\mathrm{Cov}\left(Y_i, \varepsilon_{i1}\right) = \beta_1 \mathrm{Var}\left(\varepsilon_{i1}\right)$ and hence derive expression (14). .

Solution: Substitute for Y_i from (13) into the covariance:

$$\begin{aligned}
\mathrm{Cov}\left(Y_i, \varepsilon_{i1}\right) &= \mathrm{Cov}\left(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i, \varepsilon_{i1}\right) \\
&= \mathrm{Cov}\left(\beta_0, \varepsilon_{i1}\right) + \beta_1 \mathrm{Cov}\left(X_{i1}, \varepsilon_{i1}\right) + \beta_2 \mathrm{Cov}\left(X_{i2}, \varepsilon_{i1}\right) + \mathrm{Cov}\left(u_i, \varepsilon_{i1}\right)
\end{aligned}$$

The first term is zero by definition. From part (b) we know that the second term can be expressed as $\beta_1 \mathrm{Cov}\left(X_{i1}, \varepsilon_{i1}\right) = \beta_1 \mathrm{Var}\left(\varepsilon_{i1}\right)$. The third and the fourth terms are

zero, which we have shown in (a). We then conclude that $\text{Cov}(Y_i, \varepsilon_{i1}) = \beta_1 \text{Var}(\varepsilon_{i1})$. Therefore, we obtain the required expression for β_1 as the partial effect of X_{i1} on Y_i:

$$\beta_1 = \frac{\text{Cov}(Y_i, \varepsilon_{i1})}{\text{Var}(\varepsilon_{i1})}.$$

Another way to interpret β_1 in multivariate regression (13), is to say that it is equal to the slope coefficient in a bivariate regression of y_i on ε_{i1}, namely

$$Y_i = \pi_0 + \pi_1 \varepsilon_{i1} + v_i \tag{17}$$

where $\pi_1 = \beta_1$. We say that the effect of X_{i2} has been partialled out.

(e) Under what condition will the partial effect β_1 from multivariate regression (13) be the same as the effect of X_{i1} on Y_i in a bivariate regression

$$Y_i = \gamma_0 + \gamma_1 X_{i1} + v_i. \tag{18}$$

Solution: Intuitively, $\beta_1 = \gamma_1$ when X_{i1} and X_{i2} are uncorrelated. This is the case when X_{i2} has no explanatory power for X_{i1} and hence in regression (15), $\delta_1 = 0$. This means that the residual ε_{i1} does not dependent of X_{i2}, and is a one-to-one correspondence with X_{i1}. From (16) observe that $\varepsilon_{i1} = X_{i1} - \delta_0$. Since the population intercept $\delta_0 = \mathbb{E}[X_{i1}] = \mu_{x_1}$, it is clear that $\varepsilon_{i1} = X_{i1} - \mu_{x_1}$. The residual is the demeaned variable X_{i1}. Furthermore, the result from (b) implies that $\text{Var}(\varepsilon_{i1}) = \text{Var}(X_{i1})$, and (c) implies $\text{Cov}(Y_i, \varepsilon_{i1}) = \text{Cov}(Y_i, X_{i1})$. Apply these in (**??**) to obtain

$$\beta_1 = \frac{\text{Cov}(Y_i, \varepsilon_{i1})}{\text{Var}(\varepsilon_{i1})} = \frac{\text{Cov}(Y_i, X_{i1})}{\text{Var}(X_{i1})} = \gamma_1. \tag{19}$$

For a multivariate regression (8), this means that if all the regressors $X_{i1}, \ldots, X_{ik}$ are mutually uncorrelated, $\text{Cov}(X_{ij}, X_{ik})$ for all $j \neq k$, the slope coefficients β_j, $j = 1, \ldots, k$, will be the same as the coefficients from the respective bivariate regressions of Y_i on the individual regressors X_{ij}, namely $Y_i = \beta_0 + \beta_j X_{ij} + v_i$.

(f) Suppose that instead of the true model in (13), a bivariate regression of Y_i on X_{i1} is estimated instead, so that X_{i2} is an omitted variable:

$$Y_i = \gamma_0 + \gamma_1 X_{i1} + e_i, \quad i = 1, \ldots, n. \tag{20}$$

Express the population coefficient γ_1 in terms of β_1 and β_2 from the original regression in (13), as well as the coefficient δ_1 from the regression of X_{i1} on X_{i2} given in specification (15). Hence, deduce the size of the omitted variable bias. .

Solution: Use the expression for the population coefficient γ_1 and substitute for Y_i from the true model in (13):

$$\begin{aligned}
\gamma_1 &= \frac{\text{Cov}(X_{i1}, Y_i)}{\text{Var}(X_{i1})} \\
&= \frac{\text{Cov}(X_{1i}, \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i)}{\text{Var}(X_{i1})} \\
&= \frac{\text{Cov}(X_{i1}, \beta_0)}{\text{Var}(X_{i1})} + \beta_1 \left[\frac{\text{Cov}(X_{i1}, X_{i1})}{\text{Var}(X_{i1})}\right] + \beta_2 \left[\frac{\text{Cov}(X_{i1}, X_{i2})}{\text{Var}(X_{i1})}\right] + \frac{\text{Cov}(X_{i1}, u_i)}{\text{Var}(X_{i1})}.
\end{aligned}$$

The first term is zero since β_0 is a constant. The square bracket part in the second term cancels out since $\text{Cov}(X_{i1}, X_{i1}) = \text{Var}(X_{i1})$. The square bracket in the third term is equal to δ_1, the slope coefficient from the regression of X_{i1} on X_{i2} in model (15). The fourth term is zero by assumption of exogeneity of X_{i1} in the original regression (13), that is $\text{Cov}(X_{i1}, u_i) = 0$. Putting these together, conclude that

$$\gamma_1 = \beta_1 + \beta_2 \delta_1. \tag{21}$$

This equation suggests that if X_{i2} is incorrectly omitted from the estimated regression, then the coefficient on X_{i1} will be biased by the amount equal to $\beta_2\delta_1$. The bias is the interaction of β_2, the partial effect of X_{i2} on Y_i, and δ_1, the partial effect of X_{i2} on X_{i1}. The sign of the bias will be determined by the signs of $\text{Cov}(X_{i2}, Y_i)$ and $\text{Cov}(X_{i2}, X_{i1})$. Finally, the bias is zero only if $\beta_2 = 0$ and X_{i2} contains zero explanatory power for Y_i, in which case it was correctly omitted from the original regression, or if $\delta_1 = 0$ which is equivalent to saying that $\text{Cov}(X_{i1}, X_{i2}) = 0$ and there is no linear relationship between X_1 and X_2.

S2.4 Sums of squared residuals encounters of the chi-squared kind.
Consider the following population regression of Y_i on X_i

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \ldots, n, \tag{22}$$

where β_0, β_1 are the population parameters that are known, and the error terms are given by $u_i \sim \text{iid}\, N\left(0, \sigma^2\right)$, $i = 1, \ldots, n$. For simplicity of exposition, assume that X_i are fixed numbers. Suppose you have a random sample (Y_i, X_i) for $i = 1, \ldots, n$ observations.

(a) Show that $\mathbb{E}\left[u_i^2\right] = \sigma^2$.
Solution: Since X_i are fixed numbers, we don't need to use the law of iterated expectations, like in S2.1, and can compute the unconditional expectation directly. A straightforward application of the variance formula with $\mathbb{E}[u_i] = 0$

$$\sigma^2 = \text{Var}(u_i) = \mathbb{E}\left[u_i^2\right] - (\mathbb{E}[u_i])^2 = \mathbb{E}\left[u_i^2\right]. \tag{23}$$

(b) Construct an estimator for the error variance σ^2 using the data on (Y_i, X_i), $i = 1, \ldots, n$. Call this estimator S^2. What is the relationship between S^2 and the sum of squared residuals (SSR)?
Solution: Since the population parameters are known, and need not be estimated from the data, the estimated residuals $\hat{u}_i$ are given by

$$\hat{u}_i = Y_i - \beta_0 - \beta_1 X_i, \quad i = 1, \ldots, n. \tag{24}$$

We want to estimate $\text{Var}(u_i) = \sigma^2$ from the data using expression (25). We know that since $\sigma^2 = \mathbb{E}\left[u_i^2\right]$, for the estimator of σ^2 we can use the sample analogue of $\mathbb{E}\left[u_i^2\right]$:

$$S^2 = \frac{1}{n}\sum_{i=1}^{n} \hat{u}_i^2 = \frac{1}{n}\sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_i\right)^2. \tag{25}$$

Note that we divide by n since the parameters of the model are known and we do not lose any degrees of freedom through estimation. The variance estimator S^2 is a

measure of the average variation of the residuals (deviation from the mean, where the mean is zero by construction). Recall that we refer to S as the regression standard error. The sum of squared residuals (SSR) is the total variation of the residuals:

$$SSR = \sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2 = nS^2. \tag{26}$$

(c) Use S^2 to construct a test statistic that can be used to test the hypothesis $H_0 \colon \sigma^2 = \sigma_0^2$ against $H_1 \colon \sigma^2 > \sigma_0^2$, where σ_0^2 is the hypothesized value of the error variance. Express your statistic in terms of SSR and state its distribution.

Solution: We want to use the chi-squared test for the population variance σ^2, based on the following test statistic under the assumption that the null hypothesis is true:

$$V = \frac{nS^2}{\sigma_0^2} = \frac{SSR}{\sigma_0^2} \sim \chi_n^2. \tag{27}$$

Once again, no degrees of freedom were lost as we did not have to estimate any parameters of the model. Reject H_0 at α if the computed observed sample test statistic $v > \chi^2_{1-\alpha,n}$, where $\chi^2_{1-\alpha,n}$ is the $1-\alpha$ critical value of the χ_n^2 distribution.

(d) How would your answer to (c) change if β_0, β_1 were not known and instead estimated as $\hat{\beta}_0$ and $\hat{\beta}_1$ from the data using OLS?

Solution: Recall that by estimating two parameters β_0, β_1 from the data, two degrees of freedom are lost. The estimated residuals are then given by

$$\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i, \quad i = 1, \ldots, n. \tag{28}$$

The corresponding unbiased estimator of the error variance is

$$S^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i^2 = \frac{1}{n-2} \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \right)^2 = \frac{SSR}{n-2}. \tag{29}$$

Hence $SSR = (n-2)\, S^2$. The corresponding test statistic under $H_0 \colon \sigma^2 = \sigma_0^2$ is

$$V = \frac{(n-2)\, S^2}{\sigma_0^2} = \frac{SSR}{\sigma_0^2} \sim \chi^2_{n-2}. \tag{30}$$

The main idea here is that by estimating two population parameters from the data we effectively lose two degrees of freedom. Consequently, the variance estimator (scaled appropriately) now has the χ^2_{n-2} distribution. Compare (27) and (30). Both expressions have the random variable SSR/σ_0^2 on the right-hand side. In both cases this random variable is the sum of squared residuals for the regression scaled by the population variance (assumed true under the null). The limiting distribution in both cases is χ^2, but the number of degrees of freedom is reduced by the number of parameters (regression coefficients) estimated using OLS.

(e) Finally, consider a multivariate population regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + u_i, \tag{31}$$

with k explanatory variables $X_{i1}, \ldots, X_{ik}$. The $k+1$ parameters $\beta_0, \ldots, \beta_k$ are estimated with OLS. How would you test $H_0\colon \sigma^2 = \sigma_0^2$ in this case?

Solution: The idea is exactly the same as before, but now we lose $k+1$ degrees of freedom by estimating $k+1$ population parameters, i.e. k slope parameters and one intercept. Denote the OLS estimators by $\hat{\beta}_0, \ldots, \hat{\beta}_k$, respectively. The corresponding residuals and the error variance estimator are given by

$$\begin{aligned}
\hat{u}_i &= y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \cdots - \hat{\beta}_k X_{ik} \\
S^2 &= \frac{1}{n-k-1} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \cdots - \hat{\beta}_k X_{ik} \right)^2
\end{aligned}$$

so that $SSR = (n-k-1)\, S^2$. The test statistic under $H_0\colon \sigma^2 = \sigma_0^2$ is then

$$V = \frac{(n-k-1)\, S^2}{\sigma_0^2} = \frac{SSR}{\sigma_0^2} \sim \chi^2_{n-k-1}. \tag{32}$$

This expression will be essential for constructing the F-test statistic for testing joint hypotheses. We will return to it in the next supervision.

S2.5 You better get the F out when a joint hypothesis test comes knocking. Consider the following general form of a multivariate regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + u_i, \quad i = 1, \ldots, n. \tag{33}$$

where residuals u_i are iid random variables with zero mean and constant variance. It is proposed to test a joint hypothesis $H_0^{1,2}\colon \beta_1 = \beta_2 = 0$ against the alternative $H_1^{1,2}\colon \beta_1 \neq 0$ and/or $\beta_2 \neq 0$ by using a *one at a time* testing method, that is sequentially testing the individual hypotheses on $H_0^1\colon \beta_1 = 0$ and $H_0^2\colon \beta_2 = 0$, each at the 5% significance level.

(a) Assuming that the OLS estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ are independent, show that by using this one at a time procedure, the probability of incorrectly rejecting the joint hypothesis is greater than 5%. [Hint: the decision rule for the one at a time test is: reject H_0 at $\alpha = 0.05$ if $|T_1| > 1.96$ and/or $|T_2| > 1.96$, where T_1 and T_2 are the independent t-test statistics for the individual hypotheses on $\beta_1 = 0$ and $\beta_2 = 0$, respectively. Compute the probability $\mathbb{P}\left(\{|T_1| > 1.96\} \text{ and/or } |T_2| > 1.96\right)$.]

Solution: Recall that at the significance level $\alpha = 0.05$, the probability of rejecting the null for individual coefficients is the nominal size of the test:

$$\begin{aligned}
\mathbb{P}\left(\text{Reject } H_0^1\right) &= \mathbb{P}\left(|T_1| > 1.96\right) = 0.05 \\
\mathbb{P}\left(\text{Reject } H_0^2\right) &= \mathbb{P}\left(|T_2| > 1.96\right) = 0.05
\end{aligned}$$

where the individual t-test statistics are given by $T_1 = \hat{\beta}_1 / s_{\hat{\beta}_1}$ and $T_2 = \hat{\beta}_2 / s_{\hat{\beta}_2}$. Note that T_1 and T_2 are independent random variables. Use these in the computation of $\mathbb{P}(\text{Reject } H_0^{1,2})$ for the *one at a time* procedure, where we use the fact that $|T_1| > 1.96$ and $|T_2| > 1.96$ are disjoint events:

$$\begin{aligned}
\mathbb{P}\left(\text{Reject } H_0^{1,2}\right) &= \mathbb{P}\left(\{|T_1| > 1.96\} \text{ and/or } \{|T_2| > 1.96\}\right) \\
&= \mathbb{P}\left(|T_1| > 1.96,\, |T_2| > 1.96\right) + \mathbb{P}\left(|T_1| > 1.96, |T_2| \leq 1.96\right) \\
&\quad + \mathbb{P}\left(|T_1| \leq 1.96,\, |T_2| > 1.96\right)
\end{aligned}$$

Since T_1 and T_2 are independent, this can be simplified as

$$\begin{aligned}\mathbb{P}\left(\text{Reject } H_0^{1,2}\right) &= \mathbb{P}\left(|T_1| > 1.96\right)\mathbb{P}\left(|T_2| > 1.96\right) + \mathbb{P}\left(|T_1| > 1.96\right)\mathbb{P}\left(|T_2| \leq 1.96\right) \\ &\quad + \mathbb{P}\left(|T_1| \leq 1.96\right)\mathbb{P}\left(|T_2| > 1.96\right) \\ &= 0.05 \times 0.05 + 0.05 \times 0.95 + 0.95 \times 0.05 = 0.0975\end{aligned}$$

Therefore, $\mathbb{P}(\text{Reject } H_0^{1,2}) = 0.097$, which is much higher than the size of $\alpha = 0.05$. Even if this was not a problem, then the shape of the confidence set suggests another reason for this strategy to be problematic. Drawing a confidence interval for both parameters and extending the lines up and to the right, results in a rectangle, indicating the area where the joint hypothesis would be rejected using the t-statistic. Obviously the confidence set does not coincide with the rectangle, and there are therefore various outcomes possible under which both strategies would come to the same conclusion or different conclusions. Since the proper testing strategy involves using the F-statistic, the t-statistic could result in improper inference under circumstances where the two areas do not coincide.

(b) Describe in detail the procedure for testing $H_0^{1,2}$ such that the probability of incorrectly rejecting the joint hypothesis is equal to the size of the test, $\alpha = 0.05$.

Solution: This question is asking you to describe the F-test. Use significance level α to test the hypothesis $H_0^{1,2}\colon \beta_1 = \beta_2 = 0$ against $H_0^{1,2}\colon \beta_1 \neq 0$ and/or $\beta_2 \neq 0$. Use the F-test, where regression in (33) is the unrestricted regression and the following specification is the restricted regression:

$$y_i = \beta_0 + \beta_3 X_{i3} + \cdots + \beta_k X_{ik} + u_i. \tag{34}$$

In other words, in the restricted regression X_{i1} and X_{i2} are excluded from the regression specification. The test statistic for the F-test is given by

$$F = \frac{\left(SSR_r - SSR_u\right)/q}{SSR_u/\left(n-k-1\right)} = \frac{\left(R_u^2 - R_r^2\right)/q}{\left(1-R_u^2\right)/\left(n-k-1\right)} \sim F_{q,n-k-1}, \tag{35}$$

where SSR_r / R_r^2 and SSR_u / R_u^2 are the sum of squared residuals / R^2 for the restricted and the unrestricted models, respectively, $q = 2$ is the number of restrictions, k is the number of explanatory variables in the unrestricted model, and n is the sample size. For the general case in (35), the rejection rule is $F > F_{1-\alpha,q,n-k-1}$, where $F_{1-\alpha,q,n-k-1}$ is the critical value of the $F_{q,n-k-1}$ distribution at the $1-\alpha$ probability, that is

$$F_{1-\alpha,q,n-k-1} = F_{q,n-k-1}^{-1}\left(1-\alpha\right).$$

S2.6 The wage gap is not the same without education and work experience.

This problem illustrates the difference in interpretation of slope coefficients between bivariate and multivariate regressions. For this question you will need the `STATA` file `wage330.dta`. It provides data on the following characteristics for a sample of $n = 330$ randomly selected young workers in the US in 1987:

`wage`	the individual's wage (in \$ per hour)
`male`	a 0-1 dummy variable, 1 for males
`school`	years of schooling
`exper`	years of work experience

(a) Compute an estimate of the gender wage gap for the sample as a whole, without allowing for the effects of schooling or experience. Test whether your estimate differs significantly from zero.

Solution: Column (1) in Table 3 reports the OLS estimation results for the regression

$$\texttt{wage}_i = \beta_0 + \beta_1 \texttt{male}_i + u_i, \quad i = 1, \ldots, n. \tag{36}$$

The gender wage gap is given by the estimate $\hat{\beta}_1 = 1.356$, which means that on average men get \$1.356 more than women per hour. We want to formally test $H_0 \colon \beta_1 = 0$ against $H_1 \colon \beta_1 \neq 0$ using significance level $\alpha = 0.01$. Under the null, since the sample is large, the test statistic is given by $T_{\beta_1} = \hat{\beta}_1 / s_{\hat{\beta}_1} \overset{a}{\sim} N(0,1)$. The observed value of the test statistic is given in the Table, namely $t_{\beta_1} = 4.02$. This is greater than the critical value $z_{0.995} = 2.567$ and so the null is rejected at 1%. In fact, β_1 is significant at 0.1% as the p-value of the `STATA` output suggests. The output table was produced using the `STATA` command `esttab` for summarizing regression output in a publication-style table. You can read about how to install and use this function in this online tutorial. The details of individual regressions are explained in the solutions below.

Table 3: Output from wage regressions using `STATA`

	(1) wage	(2) wage	(3) wage
`male`	1.356^{***}	1.753^{***}	1.733^{***}
	(4.02)	(5.59)	(5.55)
`school`		0.720^{***}	0.768^{***}
		(7.88)	(8.16)
`exper`			0.150^{*}
			(1.98)
`const`	5.184^{***}	-3.462^{**}	-5.221^{***}
	(21.18)	(−3.09)	(−3.67)
N	330	330	330
R^2	0.047	0.199	0.209
F-statistic	16.17	40.67	28.67

t-statistics in parentheses

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

```
>> ssc install estout, replace //install estout package
>> eststo:  quietly reg wage male
>> eststo:  quietly reg wage male school
>> eststo:  quietly reg wage male school exper
>> esttab, r2 scalars(F)
```

(b) Compute a multiple regression explaining the wage level as a function of gender and years of schooling. Test whether your new estimate of the gender wage gap differs significantly from zero. Test whether the explanatory variables (gender and schooling) are jointly significant in explaining wages. Use the expression for the omitted variable

bias you derived earlier to explain why the estimate of the gender wage gap from (a) is different. What conditions would the data have to satisfy for the two estimates to be identical? Do the two estimates measure the same underlying effect?

Solution: Columns (2) in Table 3 reports the OLS estimates of the following model

$$\texttt{wage}_i = \gamma_0 + \gamma_1 \texttt{male}_i + \gamma_2 \texttt{school}_i + v_i, \quad i = 1, \ldots, n. \tag{37}$$

Note that the estimate of the wage gap in column (2) is higher than that from column (1), $\hat{\gamma}_1 = 1.753 > 1.356 = \hat{\beta}_1$, and is more significance with a t-statistic of $t_{\hat{\gamma}_1} = 5.59$. This difference occurs because γ_1 is the partial effect of gender on wage, keeping the years of schooling fixed, while β_1 is the biased estimate of the bivariate effect where the relevant variable $\texttt{school}_i$ are omitted. The omitted-variable bias in the estimate of β_1 is negative. This is because $\texttt{school}_i$ is negatively correlated with the variable $\texttt{male}_i$ and $\widehat{\text{Cov}}\left(\texttt{school}_i, \texttt{male}_i\right) < 0$, while $\texttt{school}_i$ has a positive effect on $\texttt{wage}_i$ and $\widehat{\text{Cov}}\left(\texttt{school}_i, \texttt{wage}_i\right) > 0$, as reported in Table 4. The sign of the bias is predicted by expression (21), since the sign of the omitted variable bias is given by

$$\widehat{\text{Cov}}\left(\texttt{school}_i, \texttt{male}_i\right) \widehat{\text{Cov}}\left(\texttt{school}_i, \texttt{wage}_i\right) < 0.$$

If $\texttt{male}_i$ and $\texttt{school}_i$ were orthogonal (linearly independent), then we would have obtained the same estimates for the wage premium in (1) and (2), that is $\hat{\beta}_1 = \hat{\gamma}_1$. The same would have occurred if $\texttt{school}_i$ was irrelevant for explaining wages. In our case, $\texttt{school}_i$ is significant for explaining $\texttt{wage}_i$, as we can see from the corresponding t-statistic, $t_{\hat{\gamma}_2} = 7.88$. Finally, in order to test the overall regression significance, formulate the following joint hypothesis:

$$\begin{aligned} &H_0 \colon \gamma_1 = \gamma_2 = 0, \\ &H_1 \colon \left(\gamma_1 \neq 0\right) \cup \left(\gamma_2 \neq 0\right). \end{aligned}$$

The test is formally done using the F-test statistic, given in (35), where the unrestricted regression is given by (37) and the restricted regression is given $\texttt{wage}_i = \gamma_0 + \varepsilon_i$. Since $n = 330$, the number of estimated parameters in the unrestricted model is $k + 1 = 3$, and the number of restrictions is $q = 2$. The F-test statistic has distribution $F_{2,326}$. The output Table shows that the observed test statistic is $F = 40.67$, and the critical value for $\alpha = 0.01$ is given by $F_{0.99,2,326} = 4.671$. Hence, we reject the null hypothesis and conclude that the variables are jointly significant.

Table 4: Correlation matrix

	wage	male	school	exper
wage	1.0000			
male	0.2168	1.0000		
school	0.3502	−0.1608	1.0000	
exper	0.0115	0.0729	−0.2613	1.0000

(c) Answer the same question for a multiple regression explaining the wage level as a function of gender, years of schooling and years of experience.

Solution: We estimate the following regression:

$$\texttt{wage}_i = \pi_0 + \pi_1 \texttt{male}_i + \pi_2 \texttt{school}_i + \pi_3 \texttt{exper}_i + v_i, \quad i = 1, \ldots, n. \tag{38}$$

The output from STATA estimation is presented in column (3) of Table 3. Notice that $\texttt{exper}_i$ has a positive and a significant (at the 5% level) effect on wage, $\hat{\pi}_3 = 0.150$ and $t_{\hat{\pi}_1} = 1.98$. This means that experience is a relevant explanatory variable, even though much less important than gender or schooling. In any case, we have sufficient evidence to suggest that experience has been incorrectly omitted from the previous two model specifications. Moreover, from Table 4, we see that $\widehat{\text{Cov}}\left(\texttt{exper}_i, \texttt{male}_i\right) > 0$ and $\widehat{\text{Cov}}\left(\texttt{exper}_i, \texttt{school}_i\right) < 0$. Since the partial effect of experience on wage is positive, $\hat{\pi}_3 > 0$, these covariances applied in the expression for the omitted variable bias in (21) suggest that in regression (37), the omitted variable bias of the partial effect of gender should be positive, while the bias of the partial effect of schooling should be negative. This is exactly what we observe in the estimated output by comparing columns (2) and (3), namely $\hat{\gamma}_1 > \hat{\pi}_1$ and $\hat{\gamma}_2 < \hat{\pi}_2$. Finally, test the overall regression significance using the following hypothesis:

$$\begin{aligned} H_0 &: \pi_1 = \pi_2 = \pi_3 = 0, \\ H_1 &: \left(\pi_1 \neq 0\right) \cup \left(\pi_2 \neq 0\right) \cup \left(\pi_3 \neq 0\right). \end{aligned}$$

The F-test statistic has distribution $F_{3,326}$. Table 3 reports that the observed test statistic is $F = 28.67$. The critical value for the limiting distribution at significance $\alpha = 0.01$ is given by $F_{0.99,3,326} = 3.842$. Therefore, we reject the null hypothesis and conclude that the variables are jointly significant.

To sum up, regression (36) with $\texttt{male}_i$ only tells us the wage premium for being male without taking into account other variables. Regression (38) tells us the wage premium for being male taking into account education and experience, two variable that theoretically (and as it turns out from the estimation, also empirically) should have a significant positive effect on wage. Regression (38) tells us what the male wage premium is if we hold the other variables fixed. Explicitly, column (3) suggests that the hourly wage for a male with some fixed level of education $\texttt{school}_i = a_1$, and experience $\texttt{exper}_i = a_2$, for any values of a_1 and a_2, is $\hat{\pi}_1 = 1.733$ higher that the wage for a female with the same levels of education and experience

$$\begin{aligned} \hat{\pi}_1 = &\ \mathbb{E}\left[\widehat{\texttt{wage}}_i \,|\, \texttt{male}_i = 1, \texttt{school}_i = a_1, \texttt{exper}_i = a_2\right] \\ &- \mathbb{E}\left[\widehat{\texttt{wage}}_i \,|\, \texttt{male}_i = 0, \texttt{school}_i = a_1, \texttt{exper}_i = a_2\right] \end{aligned}$$

The coefficients in (36) and (38) have different interpretations. If $\texttt{male}_i$ is trying to measure an underlying causal effect of gender on wages, then both coefficient are trying to measure the same effect. The coefficient changes when additional controls are included. It is evident that the first estimate, bivariate effect $\hat{\beta}_1$ suffers from omitted variable bias. We cannot know with certainty whether the second estimate, partial effect $\hat{\pi}_1$, is biased or not, as we cannot tell from these results alone whether any additional covariates have been omitted in the multivariate specification (38).

S2.7 Dummy Wars. Episode V: The education strikes back.

A researcher is interested in studying the factors that determine the quality of UK secondary schools. The following regression model is estimated:

$$\texttt{ps}_i = \beta_0 + \beta_1 \texttt{exp}_i + \beta_2 \texttt{lon}_i + u_i, \quad i = 1, \ldots, n, \tag{39}$$

where the variables are defined as: $\texttt{ps}_i$ is the performance score for school i; $\texttt{exp}_i$ is the difference between school's expenditure (in thousands of pounds per pupil) and the

national average expenditure of £5000 per pupil; $\texttt{lon}_i$ is the dummy variable equal to 1 if school i is located in London and zero otherwise. Suppose that a random sample of observations on $(\texttt{ps}_i, \texttt{exp}_i, \texttt{lon}_i)$, $i = 1, \ldots, n$ is available, and that the unobserved error term u_i satisfies $\mathbb{E}[u_i \mid \texttt{exp}_i, \texttt{lon}_i] = 0$ for all i.

(a) Interpret the coefficients β_0, β_1 and β_2.

Solution: The usual interpretation (*ceteris paribus*), where β_0 is the intercept or the average score for a non-London school with expenditure of £5000 per pupil, which corresponds to $\texttt{exp}_i = 0$; β_1 is the slope coefficients or the partial effect that an additional pound of expenditure above £5000 per pupil has on the performance score (keeping school location fixed); β_3 is the additional effect for a London school compared to a non-London school (keeping expenditure fixed). Note that $\texttt{exp}_i = 0$ corresponding to the national average expenditure of £5000 per student and so

$$\begin{aligned}
\beta_0 &= \mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i = 0, \texttt{lon}_i = 0], \\
\beta_1 &= \frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}, \\
\beta_2 &= [\texttt{ps}_i \mid \texttt{exp}_i, \texttt{lon}_i = 1] - \mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i, \texttt{lon}_i = 0].
\end{aligned}$$

You can also think of β_2 as the addition to the intercept for the London schools, while keeping expenditure at the national average of £5000:

$$\beta_2 = \mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i = 0, \texttt{lon}_i = 1] - \mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i = 0, \texttt{lon}_i = 0].$$

However, this holds for any value of $\texttt{exp}_i$, not only $\texttt{exp}_i = 0$.

(b) A researcher suggests using the product $\texttt{exp}_i \times \texttt{lon}_i$ as an extra explanatory variable. Discuss a possible rationale behind such a suggestion.

Solution: The augmented regression is now given by

$$\texttt{ps}_i = \beta_0 + \beta_1 \texttt{exp}_i + \beta_2 \texttt{lon}_i + \beta_3 \texttt{exp}_i \times \texttt{lon}_i + u_i \tag{40}$$

and β_3 is an additional effect of expenditure for London schools, which are presumably richer, compared to schools outside London. In this regression, the expenditure effect for a non-London school is given by β_1, while the expenditure effect for a London school is $\beta_1 + \beta_3$. This can obtained by differentiating the new augmented regression in (40) with respect to $\texttt{exp}_i$ separately for London and non-London schools:

$$\begin{aligned}
\left.\frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\right|_{\texttt{lon}_i = 0} &= \beta_1 \\
\left.\frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\right|_{\texttt{lon}_i = 1} &= \beta_1 + \beta_3
\end{aligned}$$

Another useful way to write the same thing is:

$$\begin{aligned}
\mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i, \texttt{lon}_i = 0] &= \beta_0 + \beta_1 \texttt{exp}_i, \\
\mathbb{E}[\texttt{ps}_i \mid \texttt{exp}_i, \texttt{lon}_i = 1] &= (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\,\texttt{exp}_i.
\end{aligned}$$

Regression (40) therefore allows for two distinct London-related effects: one for the additional level effect, β_2, and one for the additional slope effect, β_3.

(c) Another researcher suggests including additive and multiplicative dummy variables interacted with `exp` for twelve different geographical regions of UK. Write down the corresponding regression and interpret the coefficients for the additive and the multiplicative dummy variables.

Solution: Denote the dummy variables for region $j = 1, \ldots, 12$ as

$$\texttt{reg_j}_i = \begin{cases} 1 & \text{school } i \text{ is in region } j \\ 0 & \text{otherwise} \end{cases}$$

where one of the twelve regions is presumably London. Let's say this is region 12 and so denote $\texttt{reg_12}_i$ to replace the original $\texttt{lon}_i$ variable. Use London as the default category in the regression, so that the coefficients will be interpreted in comparison to the effects in London. The new regression can be written as

$$\texttt{ps}_i = \beta_0 + \beta_1 \texttt{exp}_i + \sum_{j=1}^{11} \delta_j \texttt{reg_j}_i + \sum_{j=1}^{11} \gamma_j \left(\texttt{exp}_i \times \texttt{reg_j}_i\right) + u_i. \tag{41}$$

We can interpret β_0 and β_1 as the level and the slope effects for London:

$$\beta_0 = \mathbb{E}\Big[\texttt{ps}_i \,|\, \texttt{exp}_i = 0, \texttt{reg_1} = 0, \texttt{reg_2} = 0, \ldots, \texttt{reg_11} = 0\Big] \tag{42}$$

$$\beta_1 = \frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\Big|_{\texttt{reg_1}=0,\texttt{reg_2}=0,\ldots,\texttt{reg_11}=0} \tag{43}$$

The coefficient β_0 corresponds to the level effect for London, that is when all dummy variables representing UK regions take the value of 0, $\texttt{reg_1} = 0$, $\texttt{reg_2} = 0, \ldots,$ $\texttt{reg_11} = 0$ for the national average level of expenditure per student, $\texttt{exp}_i = 0$. Consequently, δ_j is the additional level effect for region j, compared to London for any level of expenditure $\texttt{exp}_i$. For example, δ_1 is the additional level effect for region 1 compared to London, for any fixed level of $\texttt{exp}_i$:

$$\begin{aligned} \delta_1 = {} & \mathbb{E}\Big[\texttt{ps}_i \,|\, \texttt{exp}_i, \texttt{reg_1} = 1, \texttt{reg_2} = 0, \ldots, \texttt{reg_11} = 0\Big] \\ & - \mathbb{E}\Big[\texttt{ps}_i \,|\, \texttt{exp}_i, \texttt{reg_1} = 0, \texttt{reg_2} = 0, \ldots, \texttt{reg_11} = 0\Big] \end{aligned}$$

Note that the overall level effect for region 1 is given by $\beta_0 + \delta_1$ and compare this with the level effect for London from (42), β_0. The coefficients for the interactive dummies, γ_j, are the additional expenditure effect for region j compared to the expenditure effect in London. For example, for region 1

$$\gamma_1 = \left(\frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\Big|_{\texttt{reg_1}=1,\texttt{reg_2}=0,\ldots,\texttt{reg_11}=0}\right) - \left(\frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\Big|_{\texttt{reg_1}=0,\texttt{reg_2}=0,\ldots,\texttt{reg_11}=0}\right).$$

You can also see this by looking at the overall partial effect of `exp` for region 1

$$\frac{\partial \texttt{ps}_i}{\partial \texttt{exp}_i}\Big|_{\texttt{reg_1}=1,\texttt{reg_2}=0,\ldots,\texttt{reg_11}=0} = \beta_1 + \gamma_1$$

and comparing it with the effect for London from (43), which is just β_1.

(d) Discuss why you may prefer adjusted R^2 to the regular R^2 as a measure of the goodness of fit regression in your regression with multiple dummy variables.

Solution: This regression effectively allows us to capture richer variability in the data and be able to explain school performance better by differentiating between varying effects by region. This means that with the addition of new relevant explanatory variables (those that have some explanatory power for the dependent vari

able), the measure of the goodness of fit, R^2 should go up. However, R^2 inevitably goes up whenever a new relevant variable is added to the right-hand side. The reason for that is that the explained sum of squares (SSE) can never decrease as a result of an addition of an extra variable in the regression. Consequently, instead of using R^2 to compare the predictive power of models with different numbers of explanatory variables, we would like to use $\bar{R}^2$, the adjusted R^2 measure, which makes an adjustment for the number of explanatory variables. The formula for R^2 is the following:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}, \tag{44}$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 : \text{Total sum of squares} \tag{45}$$

$$SSE = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 : \text{Explained sum of squares} \tag{46}$$

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 : \text{Residual sum of squares} \tag{47}$$

Note that y_i are the original observations on the dependent variable, $\bar{y}$ is the sample mean of y_i, and $\hat{y}_i$ are predicted or fitted values of y from the linear regression. The formula for the adjusted R^2, or $\bar{R}^2$, is

$$\bar{R}^2 = 1 - \left(1 - R^2\right) \frac{n-1}{n-k-1} = 1 - \frac{SSR/\left(n-k-1\right)}{SST/\left(n-1\right)}, \tag{48}$$

where n is the number of observations and k is the number of explanatory variables on the right-hand side of the regression. Note that k is the number of variables, not the number of coefficients, which in turn is $k+1$ as the intercept is included. That means that the original regression in (a) has $k = 2$ and the last regression in (c) has $k = 2 + 11 + 11 = 24$.